NoisePrivacyMap<L2Distance<RBig>, PrivacyCurveDP> for ZExpFamily<2>

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This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of the implementation of NoisePrivacyMap for `ZExpFamily<2>` with output measure PrivacyCurveDP.

1 Hoare Triple

Precondition

Compiler-verified

`NoisePrivacyMap` is parameterized as follows:

- `MI` is `L2Distance<RBig>`.
- `MO` is `PrivacyCurveDP`.

Caller-verified

None.

Pseudocode

```python
class ZExpFamily2:
    def noise_privacy_map(
        self, _input_metric: L2Distance[RBig], _output_measure: PrivacyCurveDP
    ) -> PrivacyMap[L2Distance[RBig], PrivacyCurveDP]:
        scale = self.scale
        if scale < RBig.ZERO:  #
            raise "scale must be non-negative"

        def privacy_map(d_in: RBig):
            if d_in < RBig.ZERO:  #
                raise "sensitivity must be non-negative"
            if d_in == RBig.ZERO:  #
                return PrivacyCurve().with_zCDP(0.0)
            if scale == RBig.ZERO:  #
                return PrivacyCurve().with_zCDP(float("inf"))

            rho = f64.inf_cast((d_in / scale) ** 2 / RBig(2))  #
            return PrivacyCurve().with_zCDP(rho)  #

        return PrivacyMap.new_fallible(privacy_map)
```

Postcondition

Theorem 1.1. Given a distribution `self`, the implementation returns an error if `self` is invalid. Otherwise it returns a privacy map for the mechanism `function(x) = x + Z`, where the coordinates of Z are independent discrete Gaussian samples represented by `self`, under `PrivacyCurveDP`.

Proof. Line 6 rejects a negative scale, which does not represent a valid discrete Gaussian distribution. Line 10 rejects a negative distance bound.

When `d_in` is zero, adjacent inputs are identical and line 12 returns the identity privacy curve. When the sensitivity is positive and the scale is zero, the mechanism has no finite privacy guarantee; line 14 returns the non-private curve represented by $\rho = \infty$.

It remains to consider positive rational sensitivity s and scale σ. By Theorem 14 of [CKS20], the discrete Gaussian mechanism is ρ-zCDP for

$$\rho = \frac{1}{2}\left(\frac{s}{\sigma}\right)^2.$$

Line 17 performs the arithmetic exactly over rationals and then casts toward positive infinity, so the returned floating-point value $\widehat{\rho}$ satisfies $\widehat{\rho} \geq \rho$.

Line 18 constructs a `PrivacyCurve` from this zCDP guarantee. Its privacy-profile and tradeoff queries are conservative conversions of the same zCDP guarantee. Therefore it is a valid `PrivacyCurveDP` distance for the mechanism. Increasing the distance to any `d_out` accepted by the measure ordering preserves the guarantee, which proves the privacy-map postcondition. ☐

References

[CKS20] Clément L. Canonne, Gautam Kamath, and Thomas Steinke. The discrete gaussian for differential privacy. *CoRR*, abs/2004.00010, 2020.